Reg. No. | | | | | | | | | |

Question Paper Code : 14267

B.E. / B.TECH DEGREE EXAMINATION, APRIL / MAY 2018

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. – Information Technology

13CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2013)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Define Straight line sequencing.
2. What is Message passing Multicomputers?
3. Write the register transfer sequence for storing a word in memory.
4. List the steps involved in executing the instruction "Store R1, X (R2)".
5. What are pipeline stalls? How the stalls can be overcome?
6. Define Instruction throughput for pipelined processors.
7. Define Hit Ratio.
8. Compare SRAM and DRAM.
9. What is program controlled I/O?
10. List the purpose of various registers available in I/O devices.

PART B – (5 × 16 = 80 Marks)

11. a) i. Discuss the non-restoring division algorithm. Simulate the same for **12 / 3**. (8 Marks)

11. a) ii. Explain the following addressing modes with an example and suggest the use of these addressing modes:

- Register Direct
- Auto Increment
- Indirect Address
- Indexed Address (8 Marks)

OR

11. b) i. Apply the Booth Multiplication Algorithm to perform the following multiplications:

- (- 12) x (-13)
- (- 5) x (+9)
- (+10) x (+11)

(8 Marks)

11. b) ii. Write in detail about instruction execution and straight line sequencing with an example. (8 Marks)

12. a) i. At some point in the execution of a program, registers R7, R5, and R6 contain the values 3000, 7500, and 2500, respectively. Show the contents of the Multiplexer and the registers RA, RB, RZ, RY, and R6 during steps 3 to 5 as the instruction **Subtract R7, R5, R6** is fetched and executed, and also during step 1 of the instruction that is fetched next. (8 Marks)

12. a) ii. Write in detail about the interface between processor and memory and the control signals associated with the instruction register. (8 Marks)

OR

12. b) i. List the sequence of actions needed to execute the following Instructions in terms of data path components

STORE R5,10 (R8)

SUB R3, R4, R5 (8 Marks)

12. b) ii. Discuss the hardwired control technique for generating the control signals with suitable diagrams. (8 Marks)

13. a) i. Draw the data hazard diagram and check whether operand forwarding mechanism is needed for the following instructions:

Add R1,R3,R2
Mul R5,R1,R4
Add R4,R6,R5
Add R4,R7,R2

(8 Marks)

13. a) ii. Explain how pipelining is done in the CISC style processor with an example: (8 Marks)

OR

13. b) i. Consider the pipeline execution of following sequence of instructions,

ADD R4,R3,R2

OR R7,R6,R5

SUBTRACT R8,R7,R4

Initially registers R2 and R3 contain 4 and 8 respectively. Registers R5 and R6 contain 128 and 2 respectively. The first instruction is fetched in cycle 1 and remaining instructions are fetched in successive cycles. Draw a diagram to show the pipelined execution of instructions assuming that the processor uses operand forwarding. Find the content of registers RY and RZ. (8 Marks)

13. b) ii. Discuss in detail about Branch Prediction Algorithms. (8 Marks)

14. a) i. Explain the use of DMA controllers in a computer system with neat diagram. (8 Marks)

14. a) ii. A block-set-associative cache consists of a total of 64 blocks, divided into 4-block sets. The main memory contains 4096 blocks, each consisting of 128 words.

A) How many bits are there in a main memory address?

B) How many bits are there in each of the TAG, SET, and WORD fields? (8 Marks)

OR

14. b) i. Explain the working of virtual-memory address translation with a neat block diagram. (8 Marks)

14. b) ii. Design a 64-megabit DRAM chip by organizing it as 8M x 8. Draw the internal organization of the chip and explain its working principle. (8 Marks)

15. a) i. Consider a task that reads characters typed on a keyboard, stores these data in the memory, and displays the same characters on a display screen. Show how program-controlled I/O helps in completing this task with suitable diagrams. (8 Marks)

15. a) ii. Explain in detail about bus arbitration when there are three bus masters. (8 Marks)

OR

15. b) i. Discuss in detail about SCSI and its features with an explanation on how it operates. (8 Marks)

15. b) ii. What are the functions of I/O interface? Explain in detail about parallel interface. (8 Marks)